Install docker compose and create project structure

Note: We'll reuse app.py, requirements.txt from Practical 13

```
priyansh@IDEAPAD-18UT:~$ sudo apt install docker-compose-plugin -y
docker compose version
[sudo] password for priyansh:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
docker-compose-plugin is already the newest version (5.1.3-1~ubuntu.24.04~no
ble).
The following packages were automatically installed and are no longer requir
ed:
  linux-cloud-tools-6.8.0-106 linux-cloud-tools-6.8.0-106-generic
  linux-tools-6.8.0-106 linux-tools-6.8.0-106-generic
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 86 not upgraded.
Docker Compose version v5.1.3
priyansh@IDEAPAD-18UT:~$ mkdir ~/compose-demo && cd ~/compose-demo
priyansh@IDEAPAD-18UT:~/compose-demo$
```

Use the given script in the corresponding file types

```
priyansh@IDEAPAD-18UT:~/compose-demo$ nano app.py
priyansh@IDEAPAD-18UT:~/compose-demo$ nano requirements.txt
priyansh@IDEAPAD-18UT:~/compose-demo$ nano docker-compose.yml
priyansh@IDEAPAD-18UT:~/compose-demo$
```

app.py - script

```
Ubuntu
GNU nano 7.2                              app.py *
from flask import Flask
import mysql.connector, os

app = Flask(__name__)

@app.route('/')
def home():
    try:
        conn = mysql.connector.connect(
            host=os.getenv('DB_HOST', 'db'),
            user=os.getenv('DB_USER', 'root'),
            password=os.getenv('DB_PASSWORD', 'rootpass'),
            database=os.getenv('DB_NAME', 'testdb')
        )
        return "<h1>Connected to MySQL!</h1>"
    except Exception as e:
        return f"<h1>App Running - DB error: {e}</h1>"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

requirements.txt

```
Ubuntu

GNU nano 7.2                    requirements.txt *
flask
mysql-connector-python
```

docker-compose.yml – script

```
Ubuntu

GNU nano 7.2                    docker-compose.yml *
version: '3.8'

services:
  web:
    build: .
    ports:
      - "5000:5000"
    environment:
      - DB_HOST=db
      - DB_USER=root
      - DB_PASSWORD=rootpass
      - DB_NAME=testdb
    depends_on:
      - db

  db:
    image: mysql:8.0
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: testdb
    volumes:
      - db_data:/var/lib/mysql
    ports:
      - "3306:3306"

volumes:
  db_data:
```

Start services, check them and logs for web service and DB

```
priyansh@IDEAPAD-18UT:~/compose-demo$ docker compose up -d           # Star
t both services
docker compose ps              # Check running services
docker compose logs web        # Logs for web service
docker compose logs db         # Logs for DB
WARN[0000] /home/priyansh/compose-demo/docker-compose.yml: the attribute `ve
rsion` is obsolete, it will be ignored, please remove it to avoid potential
confusion
[+] up 14/14
 ✔Image mysql:8.0 Pulled                                            88.2s
[+] Building 0.7s (2/2) FINISHED
 => [internal] load local bake definitions                          0.0s
 => => reading from stdin 504B                                      0.0s
[+] up 14/15l] load build definition from Dockerfile                0.1s
 ✔Image mysql:8.0        Pulled                                     88.2s
 ⠙ Image compose-demo-web Building                                   1.2s
```

Scale up the instances

```
priyansh@IDEAPAD-18UT:~/compose-demo$ docker compose up -d --scale web=3
WARN[0000] /home/priyansh/compose-demo/docker-compose.yml: the attribute `ve
rsion` is obsolete, it will be ignored, please remove it to avoid potential
confusion
[+] Building 0.5s (2/2) FINISHED
 => [internal] load local bake definitions                          0.0s
 => => reading from stdin 504B                                      0.0s
 => [internal] load build definition from Dockerfile                0.0s
 => => transferring dockerfile: 2B                                  0.0s
[+] up 0/1
 ⠙ Image compose-demo-web Building                                   1.1s
```

Stop all, then stop and remove containers and volumes

```
priyansh@IDEAPAD-18UT:~/compose-demo$ docker compose stop
WARN[0000] /home/priyansh/compose-demo/docker-compose.yml: the attribute `ve
rsion` is obsolete, it will be ignored, please remove it to avoid potential
confusion
priyansh@IDEAPAD-18UT:~/compose-demo$ docker compose down
WARN[0000] /home/priyansh/compose-demo/docker-compose.yml: the attribute `ve
rsion` is obsolete, it will be ignored, please remove it to avoid potential
confusion
priyansh@IDEAPAD-18UT:~/compose-demo$ docker compose down -v
WARN[0000] /home/priyansh/compose-demo/docker-compose.yml: the attribute `ve
rsion` is obsolete, it will be ignored, please remove it to avoid potential
confusion
priyansh@IDEAPAD-18UT:~/compose-demo$
```

Once done, type exit to exit the terminal.